



SYSTEM INTEGRATION AND ARCHITECTURE

MONTHLY PROGRESS REPORT

Student Information

DATE SUBMITTED	July 15, 2026	MPR NO:	4
SECTION	3A		
PROJECT TITLE	Development of a Web-Based Resident Management and Service Automation System for Barangay Fortune		
STUDENTS NAME	LAST NAME, FIRST NAME MI. – DESIGNATION 1. Adona, Andrei Joshua A. - Communication Leader 2. Bebar, Ivan Mark R. - Document Leader 3. De Dios, Owen Hart T. - Technical Leader 4. Magpayo, Aironn Paul M. - Project Leader 5. Siupan, Ivan S. - Quality Assurance Leader		

1. Tasks Planned for the Month

(What the group intended to accomplish)

- UI Quality of Life (QoL) Enhancements: Optimize the user interface and underlying source code to resolve logical inconsistencies, specifically addressing a critical navigation defect where administrative actions erroneously redirected users to the staff interface.
- Sidebar Notification System: Incorporate a sidebar notification feature following the recommendations of the project adviser to improve operational workflow and elevate the overall user experience (UX) for administrative and staff personnel.
- System Security & Hardening: Plan for enterprise-level defense mechanisms against common cybersecurity threats (e.g., brute-force attacks, malicious file uploads).
- Database & Queue Optimization: Redesign data deletion logic and queue logistics to prevent data loss and prevent residents from spamming document requests.

2. Tasks Accomplished

(What was completed)

- Navigation & UI Overhaul: Successfully refactored the system's navigation hierarchy by centralizing the AdminSidebar component, fixing UI corruption across 7 different admin pages.





- Cybersecurity Implementation: Installed backend rate-limiting (express-rate-limit) to block brute-force login attempts and strict upload filters (multer) to prevent malicious executable files from being uploaded to the server.
- Database Integrity (Soft Deletes): Replaced permanent account deletion logic with an "Archiving" system, ensuring that resident transaction histories remain intact for municipal auditing even after their accounts are removed.
- Queue Logistics & Anti-Spam: Engineered a backend filter that blocks residents from submitting duplicate requests if they already have an active pending document. Additionally, added a "No-Show Forfeiture" feature for staff to logically clear abandoned documents from the queue.
- Reporting & Notifications: Built a feature allowing administrators to instantly export System Audit Logs to an Excel-readable CSV format and laid the groundwork for automated email notifications.

3. Current System Status

(Brief Description of Progress)

- The Barangay E-Serbisyo system is currently stable, highly secure, and production-ready. Both the client and server environments boot successfully without runtime syntax errors. All critical capstone requirements (such as transaction logging, dynamic queueing, and role-based access control) are fully operational.



4. Problems Encountered

(If none, write "None")

Problems Encountered:

- File Syncing & Syntax Errors: Encountered critical syntax parsing errors (Unexpected token 'export') and string interpolation bugs in the backend Node.js server during the mass-implementation of the email notification and anti-spam features. Component Duplication: Identified major code redundancy in how the sidebar navigation was injected across multiple React pages, which was causing visual tearing and state bugs.

5. Actions Taken / Solutions

Syntax Bug Patches: Isolated the backend syntax issues within requests.js and staff.js. Rewrote the mismatched brackets and corrected the template string formats, safely restoring backend stability. Component Refactoring: Created a unified, reusable <AdminSidebar /> React component. We stripped the redundant HTML code from all dashboard pages and injected this single component, instantly fixing the visual layout and making future updates significantly easier.

6. Plan for Next Month

- Gathering of document templates for the system

Prepared by:

Bebar, Ivan Mark R. – Document Leader
(Name over Signature)

Date: July 15, 2026

Checked By:

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Date: _____